# Highlights

**RansKnow: An Open Dataset of Ransomware-Related YouTube Transcripts with Regex-Extracted MITRE ATT&CK Features**

Henry Kabuye, Biju Issac

- Open corpus of 1,128 ransomware-related YouTube video records from 92 channels
- 9 of 41 ransomware family aliases collide with ordinary English or business words
- Weak-label baselines reach macro-F1 0.90 on relevance and 0.38 on dominant tactic
- Softmax OOD detectors fail a sanity check that embedding distance passes at 0.94
- Corpus, extraction pipeline, audit and experiment code are released openly

# RansKnow: An Open Dataset of Ransomware-Related YouTube Transcripts with Regex-Extracted MITRE ATT&CK Features

Henry Kabuye[a], Biju Issac[a,*]

[a]*School of Computer Science, Northumbria University, Newcastle upon Tyne, NE1 8ST, United Kingdom*

**Abstract**

Security professionals share much of their ransomware knowledge in conference talks and vendor webinars, but we are not aware of an open transcript-level corpus that covers this material. We present RansKnow, an open corpus of 1,128 ransomware-related YouTube video records (1,034 with transcripts) from 92 cybersecurity channels, with MITRE ATT&CK-aligned family, tactic, tool, and platform features extracted by regular expressions. A label-quality audit found that 9 of 41 family aliases collide with ordinary words such as `play` and `hive`, which inflated the most-detected family about 3.5× before a context-based fix. The corpus has no independent human-annotated test set, so baseline, calibration, and conformal results measure agreement with the audited pipeline, not real-world accuracy. On that basis, baselines reach macro-F1 0.90 on relevance, 0.62 on platform, 0.38 on dominant tactic, and at most 0.20 on family and tool. Split conformal prediction reaches 0.91–0.93 coverage at a 90% target on the multiclass tasks. In out-of-distribution detection, softmax methods fail a sanity check against off-topic newsgroup posts (AUROC 0.62), while Mahalanobis distance on embeddings passes it (0.94 after controlling for length and formatting) but scores below chance on two setups whose out-of-distribution videos are selected for naming a family. We release the corpus, pipeline, and experiment code. The audit method applies to other security text datasets labeled by keywords or regular expressions.

*Keywords:* Ransomware, Cyber threat intelligence, MITRE ATT&CK, Weak supervision, Label-quality audit, Conformal prediction, Out-of-distribution detection

## 1. Introduction

Ransomware knowledge is scattered across incident-response writeups, conference talks, and vendor threat-intelligence reports, and little of it is structured for NLP or machine learning. Structured resources exist, including the MITRE ATT&CK framework [1] and datasets built from written threat intelligence (Section 2), but we are not aware of an open transcript-level corpus covering what security professionals say in talks, webinars, and walkthroughs.

YouTube gathers this material from incident-response vendors, security conferences, and independent researchers, whose incentives and house styles differ in a way no single vendor blog or threat feed replicates. Transcription is the first obstacle: 92.7% of this corpus needed

*Corresponding author

*Email addresses:* `h.kabuye@northumbria.ac.uk` (Henry Kabuye), `bissac@ieee.org` (Biju Issac)

automatic speech recognition rather than official captions (Section 3.3), and a single misheard term can change a jargon-based label (Section 4.2).

Labeling is the second. RansKnow's labels come from a regex pipeline, and the project has no human-annotated evaluation set held out from that pipeline (Section 4.3). Results from Section 5 through Section 7 therefore measure agreement with the audited pipeline, not verified accuracy.

This paper makes five contributions:

1. **A corpus.** RansKnow contains 1,128 video records, 1,034 with transcripts, from 92 cybersecurity YouTube channels, with MITRE ATT&CK-aligned features (Section 3). The corpus and code are at `https://github.com/Henrinnes/RansKnow`.
2. **A label-quality audit.** A dictionary-driven search for generic-word alias collisions found that 9 of 41 family aliases collide with ordinary words, which inflated the most-detected family about 3.5× before a disambiguation fix (Section 4, Table 1).
3. **Weak-label baselines.** We report results across five prediction tasks, three feature representations, and five models, with calibration and conformal prediction on top (Sections 5 to 7).
4. **Out-of-distribution results.** These scores need no correct dominant-tactic label, although two of the four setups define their out-of-distribution populations with the regex (Section 8).
5. **Counterfactual and perturbation checks.** A classifier trained on the fixed labels and scored on videos it did not train on still ranks the videos the fix removed slightly above other negatives (AUROC 0.58), a weak trace of the fragility the audit found (Section 9).

## 2. Related Work

**Ransomware research.** Recent surveys concentrate on ransomware behavior and defense: Beaman et al. [2] review advances in prevention and detection, and Oz et al. [3] survey ransomware evolution, taxonomy, and defense solutions. Meland et al. [4] study the ransomware-as-a-service economy through two years of observation of darknet markets and forums. RansKnow addresses a different question: how practitioners describe ransomware in public talks and walkthroughs, and how reliably that description can be turned into structured data.

**Mapping threat reports to ATT&CK.** A related line of work extracts structured behavior from written reports. TTPDrill [5] extracts threat actions from threat reports and aligns them with attack patterns, rcATT [6] compares classifiers for retrieving ATT&CK tactics and techniques from reports and releases a tool based on the results, and Orbinato et al. [7] report an experimental study of machine-learning classification of unstructured threat intelligence into attack techniques. These works take written text as input. RansKnow applies a deterministic regex method to mostly ASR transcripts and audits the labels that method produces.

**Ransomware and cyber-threat-intelligence datasets.** The MITRE ATT&CK framework [1] that RansKnow's tactic taxonomy is aligned to was designed for exactly this kind of cross-source mapping, and prior datasets have built structured resources against it

from written cyber-threat-intelligence (CTI) text: MalwareTextDB [8] provides span-level annotations (39 APT reports, 6,819 sentences) for malware characteristics, and the more recent AnnoCTR [9] links entities, tactics, and techniques across 400 commercial CTI reports, with human annotation throughout. Both are built from written prose – vendor blog posts and reports, produced already in structured-adjacent form for a threat-intelligence-consuming audience. RansKnow's source material is instead spoken narration, transcribed at 92.7% via ASR (Section 3.3), from a broader and more heterogeneous set of speakers (vendor analysts, conference presenters, independent researchers) than a CTI report's house style, and its structured features are produced entirely by an audited regex/keyword pipeline (Section 4) rather than human annotation – a different, explicitly weaker, point on the label-quality spectrum those two datasets occupy, traded for scale and openness (we are not aware of a comparable open, transcript-level, MITRE-aligned video corpus, although our search was not a systematic review).

**Weak and distant supervision.** Using an automated heuristic in place of human labels is a well-studied tradeoff, not a shortcut unique to this project: Mintz et al. [10] formalized distant supervision for relation extraction by aligning text against a knowledge base rather than hand-labeling it, accepting a known noise ceiling in exchange for scale, and Ratner et al. [11]'s Snorkel generalizes the idea to multiple, possibly conflicting, weak labeling functions combined probabilistically. The Knowledge Agent (Section 4) is a single deterministic labeling function rather than an ensemble of weak sources, which makes it simpler to audit (the alias-collision analysis in Section 4.1 would be considerably harder against a black-box combination of many weak labelers) but removes Snorkel-style noise-cancellation-via-redundancy as a mitigation — a tradeoff made explicit rather than left implicit, and the direct motivation for Section 4.3's scope note running through every downstream section. Label-quality research reaches similar concerns from the model side: confident learning [12] characterizes and identifies label errors in datasets, and Northcutt et al. [13] estimate an average of 3.3% label errors across the test sets of 10 widely used benchmarks and show that such errors can destabilize benchmark results. RansKnow's audit starts from the labeling function instead, searching its alias list for collisions with ordinary vocabulary.

**Conformal prediction and out-of-distribution detection.** Split conformal prediction [14, 15] and its distribution-free coverage guarantee (Section 7) have seen growing NLP and security application as a distribution-free alternative to ad hoc confidence thresholds; Angelopoulos and Bates [16] survey the modern toolkit this paper's Section 7.1 draws on directly. Out-of-distribution detection (Section 8) via softmax-confidence baselines [17], energy scores [18], and class-conditional embedding distance [19] is comparatively less explored specifically for weak-label security-NLP classifiers, where – as Section 8.2 shows concretely – the usual confidence-based methods can fail in a domain-specific way (a classifier's most confident predictions coinciding with exactly the "nothing detected" signature that off-topic input also produces).

## 3. Dataset Construction

RansKnow [20, 21] consists of 1,128 video records sourced from 92 cybersecurity YouTube channels, of which 1,034 have an extracted transcript with real content (four of these videos are listed twice, under two registry entries for the same organization, so the 1,034 transcripts

cover 1,030 distinct videos); the remaining 94[1] could not be transcribed by any method in the fetch cascade.

### *3.1. Channel selection*

Channels were curated in two batches. Batch 1 (50 channels, `C01`–`C50`) covers established DFIR vendors, security conferences, and independent researchers. Batch 2 (51 candidate channels, `C51`–`C101`) expanded coverage into threat-intelligence vendors, additional conferences, and OT/ICS-security and government channels; 42 of the 51 candidates yielded usable transcripts, 1 was a confirmed duplicate of an existing Batch 1 channel, 4 had no discoverable dedicated YouTube channel, and 4 resolved to real channels with zero ransomware-keyword-matching content in their upload history. Full channel-level provenance for both batches is in `rubrics/Dataset_Channel_Registry_Updated_50_fixed_urls.xlsx` (Batch 1) and `rubrics/Dataset_Channel_Registry_Batch2.xlsx` (Batch 2), released alongside the corpus rather than reproduced inline as a 92-row table.

### *3.2. Video selection*

Within each channel, videos were selected by scanning recent uploads for a fixed 21-term keyword list (`ransomware`, `extortion`, `lockbit`, `incident response`, ...; full list in Appendix A) matched against title and description, capped at 20 videos per channel and a minimum duration of 5 minutes (short clips and trailers excluded). This is a convenience sample, not a random or exhaustive one, and the selection mechanism itself introduces bias worth naming rather than assuming away: a video is only in scope if its *title or description* happens to contain one of the 21 terms, which conditions the corpus on each channel operator's own titling and SEO/description-writing habits. A channel that discusses ransomware extensively but titles episodes generically (e.g. by client-engagement number rather than threat name) would be under-represented regardless of its actual content relevance, and channels that lean on a small number of high-profile named families (`lockbit`, `conti`, ...) in titles for engagement will be over-represented relative to channels discussing less headline-grabbing groups. This compounds with Section 10's broader sampling-bias point: keyword-based inclusion is a second, independent bias layer on top of channel curation, not a neutral technical filtering step.

### *3.3. Transcript acquisition*

Transcripts were acquired via a three-stage cascade: (1) the YouTube Data API's official captions where available; (2) `yt-dlp`'s auto-generated captions; (3) local Whisper transcription [22] as a fallback. **This cascade matters more than a footnote**: only 74 of 1,034 transcripts (7.2%) came from official captions. 959 (92.7%) are ASR output – 554 from `whisper_base` (the smallest, least accurate Whisper checkpoint) and 405 from a local Whisper fallback, 1 from `yt-dlp` captions. This raw composition is Finding B of the label-quality audit (Section 4); the downstream implication – what an ASR-dominated corpus does to jargon-term detection specifically – is reported there rather than duplicated here.

[1]Verified directly against the released `transcripts/` tree, not assumed from 1128 − 1034: all 94 are confirmed failed transcript extraction (no method in the fetch cascade, Section 3.3, produced usable text), not videos excluded post-fetch for another reason.

## 4. Label-Quality Audit

The Knowledge Agent pipeline (`Scripts/knowledge_agent.py`) extracts structured features from every transcript via regex matching: ransomware family mentions (against a maintained alias list), MITRE ATT&CK tactic mention counts across 10 tactics, offensive tool mentions across 8 tools, and platform signals (Windows/Linux/ESXi). Three concrete problems were found and are described here rather than left as an implicit caveat, because a downstream classifier trained on these labels inherits whatever bias they contain.

### *4.1. Finding A: generic-word alias collisions*

Nine of 41 family aliases collide with common English or business words (`play`, `hive`, `cactus`, `agenda`, `clop`, `inc`, `maze`, `medusa`, `royal`) – identified programmatically via dictionary lookup against `/usr/share/dict/words`, then confirmed by manual transcript inspection (e.g. `hive` matching "registry hive," a routine DFIR term, rather than Hive ransomware). A windowed-context disambiguation fix (requiring a ransomware-context term within ∼15 tokens of the match) was applied and reduces false-positive volume substantially without eliminating it – residual false positives still occur when context vocabulary is itself dense (see `Scripts/audit_family_aliases.py` for the audit methodology and Table 1 for before/after counts). The precision and recall of the fixed labels have not been measured against human judgment, so Table 1 shows how many detections the fix removed, not how many of the remaining detections are correct.

Table 1: Before/after family mention counts for the 9 flagged aliases, windowed-context disambiguation applied. `Play` alone accounted for 386 of 465 (83%) of the family-tagged videos in the audit before the fix.

| Family | Before | After | Dropped |
|---|---|---|---|
| Play | 386 | 111 | 275 |
| Qilin | 88 | 21 | 67 |
| Hive | 23 | 10 | 13 |
| Maze | 11 | 5 | 6 |
| Royal | 28 | 19 | 9 |
| INC | 10 | 7 | 3 |
| Cactus | 9 | 6 | 3 |
| Cl0p | 11 | 11 | 0 |
| Medusa | 7 | 7 | 0 |

### *4.2. Finding B: ASR-dominated transcript quality*

As noted in Section 3.3, 92.7% of transcripts are ASR output rather than official captions. Aggregate detection-rate comparisons across `Transcript_Provider` do not show an obvious gap, which is not the same as no effect: individual mistranscriptions of rare jargon can matter without moving an aggregate rate, since a single missed `rclone` mention changes that video's `Tool_Rclone` feature from 1 to 0 regardless of how many other videos were transcribed correctly.

Source audio is not stored locally (the released dataset ships transcripts, not audio), so the originally planned manual audio-vs-transcript listening check is automated here instead: for a random sample of 40 `whisper_base`-transcribed videos (the smallest, least accurate checkpoint), the opening 6 minutes of audio was re-fetched and re-transcribed with Whisper's `small` checkpoint, treated as a stronger reference, and every occurrence of six fixed jargon terms (`LockBit`, `Mimikatz`, `PsExec`, `Cobalt Strike`, `rclone`, `BloodHound`, plus known phonetic-mangling variants) in the reference transcript was checked against the corresponding stored `whisper_base` transcript. 35 of 40 videos were successfully re-fetched and re-transcribed (5 audio-download failures); across those 35, the six jargon terms appeared at all in only 3 (video, term) instances within the sampled opening 6 minutes – jargon density is low even in ransomware-focused content – and 1 of those 3 was missed entirely by `whisper_base`: `rclone`, mentioned once in the stronger reference transcript for one video and zero times (correct spelling or a known mangled variant) anywhere in that video's stored transcript, while `bloodhound` (same video) and `cobalt strike` (18 mentions in another video) were both correctly detected. A miss rate of 1/3 from a spot-check this small (3 instances) is not a precise estimate and should not be read as one – the honest conclusion is qualitative, not quantitative: jargon-term mistranscription is a real, observed failure mode of this corpus's dominant transcription method, consistent with Section 9.3's simulated-ASR-noise sweep, and a larger, unabridged spot-check (full video length, not a 6-minute cap; larger sample) is the natural follow-up this finding motivates rather than resolves.

### *4.3. Scope: no independent ground truth*

An earlier design stage of this project planned a held-out, independently annotated evaluation subset (180 videos, stratified by channel type, year, and current family-count) specifically to break the circularity of evaluating a classifier against the same weak-label pipeline it was trained on. That subset was never annotated. Recruiting external annotators would have required ethics approval beyond the project's existing approval for public data, and was not undertaken; a single-investigator pass was considered as a fallback and rejected as still too close to the system under evaluation to serve as independent ground truth in a way worth claiming in a paper.

The consequence is stated here plainly, and repeated at the point every affected result is reported, rather than left as a single Limitations bullet a reader could miss: **every quantitative result in Section 5 onward, apart from the population-separation scores in Section 8 and the behavioral checks in Section 9, measures agreement with the audited Knowledge Agent extraction pipeline (Section 4), not verified accuracy against human judgment.** Section 4 demonstrates concretely that this pipeline's labels contain real, non-trivial error even after the disambiguation fix (Finding A); nothing downstream corrects for that. Readers should treat macro-F1 and calibration numbers in this paper as characterizing how well models reproduce a specific, imperfect, documented extraction process – not as claims about ransomware-domain understanding in an absolute sense. Section 10 returns to this as the paper's central limitation.

## 5. Task Definitions and Baselines

Five prediction tasks are defined over the corpus: ransomware family (multi-label, 18 classes after excluding 3 singleton-occurrence classes, see Table 2), dominant MITRE tactic (multi-class, 11 classes), platform signal (multi-class, 4 classes), offensive tool mentions (multi-label, 7 classes after excluding `MegaNZ`, which has zero occurrences in the corpus), and a ransomware-relevance binary screen (derived, not a stored column: `Family_Count > 0 OR Tactic_Impact > 0`).

Table 2: Task taxonomy.

| Task | Type | Classes | Label source |
|---|---|---|---|
| Family | multi-label | 18 | alias regex (§4.1) |
| Dominant tactic | multi-class (+none) | 11 | max of 10 tactic counts |
| Platform signal | multi-class (+none) | 4 | max of 3 platform counts |
| Tool mentions | multi-label | 7 | per-tool regex |
| Ransomware-relevant | binary | 2 | derived |

### *5.1. Model ladder and validation protocol*

Each task is evaluated over two feature representations (TF-IDF and mean-pooled sentence embeddings[2]), with structured Knowledge Agent columns added for `family` and `dominant_tactic` only, and five models (the rule-based Knowledge Agent itself as baseline 0, Logistic Regression, Linear SVM, Random Forest, LightGBM [24]), under three cross-validation protocols: stratified random $k$-fold, `GroupKFold` by `Channel_ID` (to detect channel-style leakage – a model can learn "this is CrowdStrike's narration style" instead of the content), and a temporal split (train pre-2024, test 2024+, since 79% of the corpus is from 2024 onward).

### *5.2. Results*

Per Section 4.3, every number in this section is weak-label agreement – how well a model reproduces the Knowledge Agent's own extraction, not verified accuracy. The rule-based Knowledge Agent baseline scores a tautological macro-F1 of 1.000 on every task (checked against its own labels by construction; included as baseline 0, not as evidence of anything) and is omitted from Table 3 accordingly. Structured features are excluded from the dominant-tactic comparison for a related reason: they contain the tactic counts its label is the argmax of. That combination scores 0.813 with LightGBM, which measures how well a tree model learns an argmax over its own inputs, so Table 3 reports the best representation without those counts.

[2]384-dim `all-MiniLM-L6-v2`, chunked at ∼200 words per chunk and mean-pooled, computed via a dedicated Python environment (`torch` 2.5.1) since the reference implementation's default environment ships an older `torch` that cannot satisfy `sentence_transformers`' [23] requirements; see `Scripts/precompute_embeddings.py`. The 20 Newsgroups embeddings and the experiments rerun for this version used Python 3.12.3 with `torch` 2.15.0, `sentence_transformers` 6.2.0, scikit-learn 1.9.1, and LightGBM 4.7.0.

Table 3: Best stratified-CV macro-F1 per task, across the eligible representations and trained models (LogReg, Linear SVM, Random Forest, GBM). Weak-label agreement, not verified accuracy (§4.3). Mean ± std is across 5 stratified-random split seeds (0–4) at the reported (representation, model) cell, model `random_state` held fixed – see the paragraph below the table.

| Task | Representation | Model | Macro-F1 (mean ± std) |
|---|---|---|---|
| Ransomware-relevant | TF-IDF | GBM | $0.897 \pm 0.006$ |
| Platform signal | TF-IDF | GBM | $0.616 \pm 0.008$ |
| Dominant tactic | TF-IDF | GBM | $0.377 \pm 0.009$ |
| Tool mentions | TF-IDF | LogReg | $0.166 \pm 0.032$ |
| Family | embedding | LogReg | $0.115 \pm 0.019$ |

The single-seed point estimates used elsewhere in this paper are each the seed-0 value of the same cell, within 0.015 of the 5-seed mean for every task except `tool`, whose seed-0 value (0.200) sits 0.034 above its mean. Std is smallest for `relevance` (0.006) and largest for `tool` (0.032, ∼20% of its own mean), consistent with `tool`'s already-noted instability across CV protocols (Table 4). Only the stratified-random split's train/test partition varies with the seed here; `channel_grouped` (`GroupKFold`) and `temporal` (a fixed pre/post-2024 cutoff) are both deterministic single partitions with no random component (`Scripts/rk_pipeline/splits.py`), so a seed sweep is not a meaningful question for those two and is not computed for them. Model `random_state` (RF/GBM) is likewise held fixed at 0 throughout: this table isolates variance from which videos land in train vs. test under 5 different stratified partitions specifically, not full stochastic-training variance, and is reported as exactly that rather than a broader claim (`outputs/experiments/phase1_multiseed.json`, `Scripts/phase1_multiseed.py`).

Two patterns stand out: (1) GBM is the strongest model on relevance, dominant tactic, and platform, and logistic regression on tool and family; (2) `family` and `tool` are weak under every combination tried (family never exceeds 0.13 macro-F1; tool peaks at 0.20 on the stratified split and reaches only 0.08 on the other two). Section 9.1's channel-identity check makes literal channel naming an unlikely explanation for the family result, leaving the weak-label ceiling (§4.3) and class sparsity (18-class family and 7-class tool taxonomies, several classes with single-digit support) as the remaining candidates, neither of which this paper isolates.

Table 3 reports only the single best (representation, model) cell per task. The full sweep has 144 trained combinations: TF-IDF and embeddings for all 5 tasks, plus structured features for family and dominant tactic, each with 4 trained models and 3 CV protocols (`outputs/experiments/phase1_*.csv`). Figure 1 summarizes it, leaving out the circular dominant-tactic structured cells and taking the best remaining representation for each (task, model, split) cell so the figure stays legible, and Table 4 gives exact numbers for the stratified-random split.

Figure 2 plots Table 4 as a 3D stem plot rather than solid bars: each (task, model) cell is a thin marked stem rather than an opaque block, so no value is hidden behind a taller neighbor from any viewing angle, which a solid-bar rendering of the same data could not guarantee.

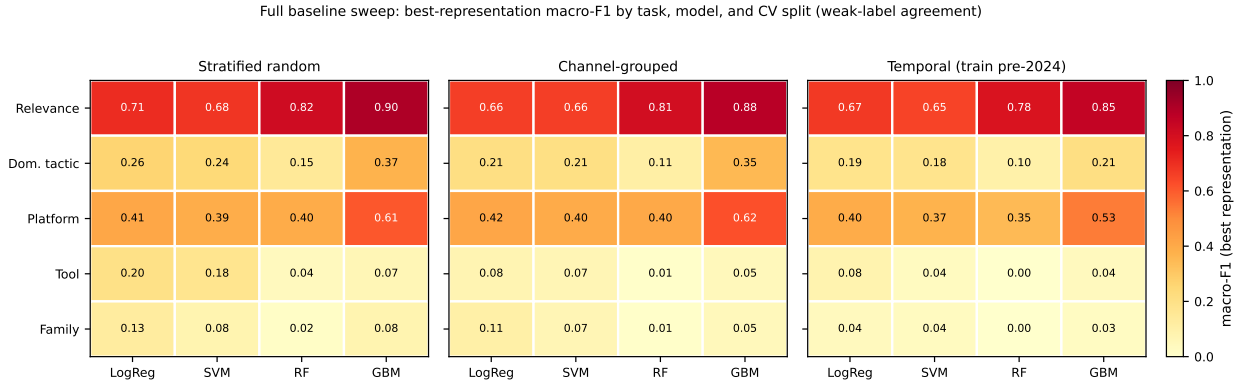



Figure 1: Best-representation macro-F1 for every (task, model) pair, one panel per CV protocol (weak-label agreement, §4.3). `family/tool` stay weak under every model and every split; `relevance` is both the strongest task and the most split-robust.

Table 4: Best-representation macro-F1 per (task, model), stratified-random split (weak-label agreement, §4.3), excluding dominant-tactic structured features. The temporal split costs some tasks far more than others, per Figure 1: the best family score falls from 0.13 to 0.04, tool from 0.20 to 0.08, and dominant tactic from 0.37 to 0.21, while platform (0.53) and relevance (0.85) hold up better.

| Task | LogReg | SVM | RF | GBM |
|---|---|---|---|---|
| Relevance | 0.71 | 0.68 | 0.82 | 0.90 |
| Dominant tactic | 0.26 | 0.25 | 0.15 | 0.37 |
| Platform | 0.41 | 0.39 | 0.40 | 0.61 |
| Tool | 0.20 | 0.18 | 0.04 | 0.07 |
| Family | 0.13 | 0.08 | 0.02 | 0.08 |

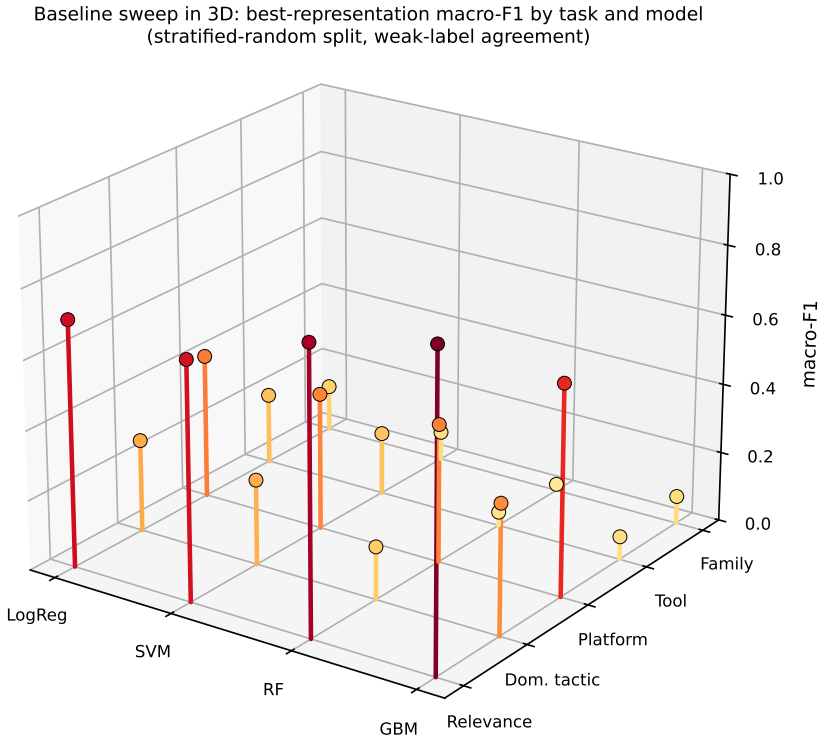



Figure 2: Table 4 as a 3D stem plot – every point visible regardless of viewing angle, unlike a solid-bar rendering of the same data.

## 6. Uncertainty Quantification

**Scope note (§4.3 applies here directly):** calibration is normally reported against ground truth – does an 80%-confidence prediction turn out correct roughly 80% of the time. Without independent labels, the only calibration target available here is the weak-label distribution itself, which changes what a well-calibrated result would mean: a model perfectly calibrated against its own training signal has learned to be a confident, well-calibrated echo of a regex, not necessarily a well-calibrated model of ransomware content. This section is retained, with that meaning made explicit at every reported number, because reliability diagrams and Expected Calibration Error against the weak labels are still informative about a narrower question worth answering: does a model's confidence track its *agreement with the extraction pipeline*, or is it uniformly overconfident regardless of agreement. The latter would be a red flag on its own.

### *6.1. Calibration against weak labels*

For a multiclass task, let $\hat{p}(\cdot \mid x)$ be the trained model's predicted class distribution for input $x$, and let $\hat{c}(x) = \arg\max_c \hat{p}(c \mid x)$ with confidence $\hat{p}_{\max}(x) = \max_c \hat{p}(c \mid x)$. Agreement with the weak label $y^{\mathrm{KA}}(x)$ (the Knowledge Agent's own extraction) is $a(x) = \mathbb{1}[\hat{c}(x) = y^{\mathrm{KA}}(x)]$. For the two multi-label tasks (`family`, `tool`), whose models are independent per-class binary classifiers (Section 5), each (sample, class) pair $(x, k)$ is treated as its own calibration point, with confidence $\hat{p}_{\max}(x, k) = \max\big(\hat{p}_k(x), 1 - \hat{p}_k(x)\big)$ and agreement against the binary weak label thresholded at 0.5.

Reliability is summarized with $B$=10 equal-width confidence bins $\{I_b\}_{b=1}^{B}$ and the standard Expected Calibration Error,

$$\mathrm{ECE} \;=\; \sum_{b=1}^{B} \frac{|S_b|}{N} \left|\mathrm{acc}(S_b) - \mathrm{conf}(S_b)\right|, \qquad S_b = \{x : \hat{p}_{\max}(x) \in I_b\}, \tag{1}$$

where $\mathrm{acc}(S_b) = \frac{1}{|S_b|}\sum_{x \in S_b} a(x)$ and $\mathrm{conf}(S_b) = \frac{1}{|S_b|}\sum_{x \in S_b} \hat{p}_{\max}(x)$. Per the scope note above, $a(x)$ measures agreement with $y^{\mathrm{KA}}$, not correctness, so Eq. (1) is a *weak-label ECE*. The headline number below the scope note is the confident-disagreement rate at threshold $\tau = 0.9$,

$$\mathrm{CDR}_\tau \;=\; \frac{1}{N}\sum_{i=1}^{N} \mathbb{1}[\hat{p}_{\max}(x_i) > \tau] \cdot \big(1 - a(x_i)\big), \tag{2}$$

the closest available analogue to a "confidently wrong" rate without independently verified labels to actually be wrong against.

Each task is evaluated on the 5-fold stratified out-of-fold predictions of its best (feature, model) combination from Table 3.

Figure 3 plots the reliability curves underlying Table 5. Two things worth naming honestly rather than smoothing over: first, per-bin counts are small (single digits to low tens per bin at $N$=1034), so individual bin heights are noisy – the curves are directionally informative, not point-estimate-precise. Dominant tactic has the largest ECE (0.178): without the tactic counts, its model is markedly overconfident against the weak labels. Second, the low per-label ECE of `tool` and `family` reflects class imbalance more than skill. Most of

Table 5: Weak-label calibration (Equations (1) and (2)) and the RF tree-vote entropy proxy (Equation (3)), best (feature, model) per task. $N$ is the number of (sample) or (sample, class) calibration points. Entropy columns give mean entropy when the forest agrees and disagrees with the weak label, and how well entropy alone separates the two (AUROC, Algorithm 1).

| Task | $N$ | ECE | $\text{CDR}_{0.9}$ | Entropy (agr./dis.) | AUROC |
|---|---|---|---|---|---|
| Relevance | 1034 | 0.079 | 0.071 | 0.910 / 0.980 | 0.763 |
| Dominant tactic | 1034 | 0.178 | 0.113 | 0.654 / 0.710 | 0.729 |
| Platform | 1034 | 0.110 | 0.104 | 0.716 / 0.770 | 0.682 |
| Tool (per-label) | 7238 | 0.016 | 0.016 | 0.121 / 0.275 | 0.977 |
| Family (per-label) | 18612 | 0.018 | 0.019 | 0.095 / – | – |

their (sample, class) pairs are negative, so predicting "absent" agrees with the weak label almost everywhere, despite the low macro-F1 in Table 3, which weights classes equally rather than by prevalence. For the same reason `family`'s RF entropy proxy, which runs on TF-IDF rather than the task's best embedding representation, has no "disagree" entries to average.

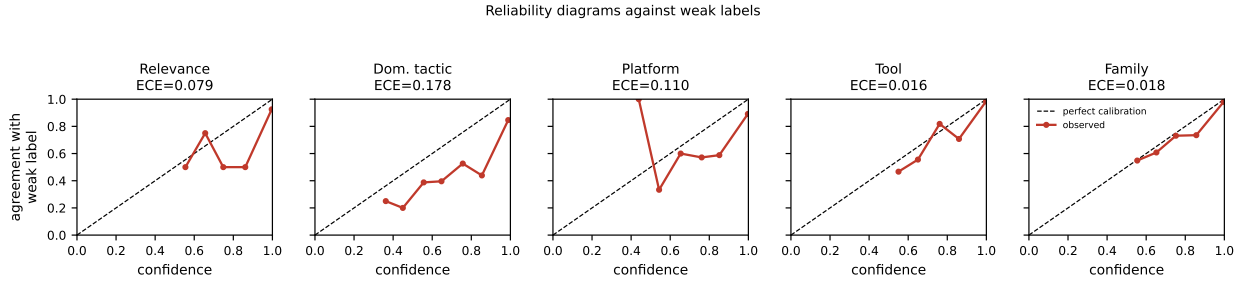



Figure 3: Reliability diagrams against weak labels, one panel per task with a usable calibration curve (Table 5). Diagonal is perfect calibration.

### *6.2. A free uncertainty proxy: Random Forest tree-vote entropy*

Reliability diagrams need a held-out calibration protocol; a Random Forest offers a per-prediction uncertainty signal with no separate calibration step. For an input $x$ and a forest of $T$ trees producing individual (hard) predictions $t_1(x), \ldots, t_T(x)$ over $K$ classes, let $\pi_c(x) = \frac{1}{T}\sum_{j=1}^{T} \mathbb{1}[t_j(x) = c]$ be the empirical vote share for class $c$. The normalized tree-vote entropy is

$$H(x) \;=\; -\frac{1}{\log K}\sum_{c=1}^{K} \pi_c(x)\,\log \pi_c(x) \;\in\; [0,1], \tag{3}$$

with $H(x)$=0 when every tree agrees and $H(x)$=1 at a uniform split across all $K$ classes. For the multi-label tasks, $K$=2 per class and $H(x,k)$ is averaged over classes $k$ for a per-sample summary. Algorithm 1 is the exact out-of-fold procedure used to produce Table 5's entropy columns.

The in-line comment in Algorithm 1 records a real implementation pitfall, not a hypothetical one: scikit-learn [25] fits each tree in a `RandomForestClassifier` on internally

**Algorithm 1** RF tree-vote entropy, 5-fold out-of-fold

**Require:** dataset $\mathcal{D}$, task with label function $y(\cdot)$, feature builder $\phi$

$\mathcal{F} \leftarrow$ 5 stratified folds of $\mathcal{D}$
**for** fold $(\mathcal{D}_{\text{train}}, \mathcal{D}_{\text{test}}) \in \mathcal{F}$ **do**
    fit $\phi$ on $\mathcal{D}_{\text{train}}$; fit Random Forest $\{t_j\}_{j=1}^{T}$ on $\phi(\mathcal{D}_{\text{train}})$
    **for** $x \in \mathcal{D}_{\text{test}}$ **do**
        $t_j(x) \leftarrow$ individual tree prediction, $j = 1, \ldots, T$ ▷ sklearn returns internal integer-encoded classes here, not the original label strings – see note below
        recover original class labels via the fitted forest's class index before comparing to $y(x)$
        compute $H(x)$ via Equation (3); record $a(x) = \mathbb{1}[\text{forest majority vote} = y(x)]$
    **end for**
**end for**
**return** $\{(H(x), a(x))\}_{x \in \mathcal{D}}$, pooled across folds

label-encoded integer targets, so `tree.predict()` returns class *indices*, not the original label strings the ensemble's own `.predict()` returns. Comparing tree-level votes directly against string labels without mapping back through the fitted forest's class index silently produces 0% apparent agreement on every multiclass task – caught in this project's own Phase 2 run (initial numbers showed exactly that pattern across all three multiclass tasks simultaneously, which is what made it legible as a bug rather than a real result) and is recorded here as a concrete instance of the caution Section 10 urges about verifying rather than assuming standard library internals. Table 5's entropy AUROC column shows the proxy works well once fixed: 0.68–0.98 across tasks, best on `tool` (many easy true negatives) and weakest on `platform`.

## 7. Conformal Prediction

**Scope note:** split conformal prediction's distribution-free coverage guarantee is a statement about exchangeability with the calibration set, not about label correctness – it holds formally even when calibrated against weak labels, because the guarantee is "the true *weak* label falls in the predicted set $(1-\alpha)$ of the time," not "the true *real-world* label does." That distinction is the entire content of this scope note: conformal coverage against the Knowledge Agent's labels is a well-defined, honestly reportable number, but it validates agreement with the extraction pipeline, not correctness. Reported as such.

### *7.1. Method: split conformal with the LAC score*

Split conformal prediction (14, 15; see 16 for a tutorial treatment) uses a nonconformity score $s(x, y)$ that is small when $y$ is a plausible label for $x$. This paper uses the least-ambiguous-set (LAC) score,

$$s(x, y) \;=\; 1 - \hat{p}(y \mid x), \tag{4}$$

calibrated on a held-out split $\mathcal{D}_{\text{calib}}$ of size $n$ disjoint from both the training and test data. The calibration threshold uses the standard finite-sample correction,

$$\hat{q} \;=\; \text{Quantile}\left( \{s(x_i, y_i)\}_{i=1}^{n} ;\; \frac{\lceil (n+1)(1-\alpha) \rceil}{n} \right), \tag{5}$$

and the prediction set for a new input $x$ is

$$C(x) \;=\; \big\{\, y \;:\; \hat{p}(y \mid x) \geq 1 - \hat{q} \,\big\}. \tag{6}$$

Under exchangeability of $\mathcal{D}_{\text{calib}}$ and the test point, this construction guarantees

$$\mathbb{P}[\, y_{\text{test}} \in C(x_{\text{test}}) \,] \;\geq\; 1 - \alpha, \tag{7}$$

with $\alpha$=0.10 (90% target coverage) throughout. As the scope note above states, $y_{\text{test}}$ here is the weak label $y^{\text{KA}}$: Equation (7) is a real, distribution-free guarantee about agreement with the extraction pipeline, not about real-world correctness. For `dominant_tactic` and `platform`, Equations (4) to (6) apply directly over the task's full class vocabulary. `family` and `tool` have no single multiclass structure to conformalize – each is an independent per-class binary classifier (Section 5) – so each class $k$ gets its own binary LAC calibration and its own $\hat{q}_k$, and a sample's per-label prediction sets are reported both individually and jointly (all labels simultaneously covered).

Algorithm 2 is run for 5 repeated random 60/20/20 train/calibration/test splits per task, results averaged across repeats rather than reported from one arbitrary split – a single 20% calibration set at this sample size ($n$≈200) is not a reliable enough estimate of $\hat{q}$ to report alone.

**Algorithm 2** Split conformal calibration and coverage evaluation

**Require:** dataset $\mathcal{D}$, task, target coverage $1-\alpha$, repeats $R$

**for** $r = 1, \ldots, R$ **do**
  split $\mathcal{D}$ into $\mathcal{D}_{\text{train}}$ (60%), $\mathcal{D}_{\text{calib}}$ (20%), $\mathcal{D}_{\text{test}}$ (20%), stratified
  fit model on $\mathcal{D}_{\text{train}}$
  compute $\hat{q}$ on $\mathcal{D}_{\text{calib}}$ via Equations (4) and (5)
  **for** $x \in \mathcal{D}_{\text{test}}$ **do**
    build $C(x)$ via Equation (6); record $\mathbb{1}[y^{\text{KA}}(x) \in C(x)]$ and $|C(x)|$
  **end for**
**end for**
pool test-set coverage indicators across all $R$ repeats
report marginal coverage, and coverage restricted to the Year≥2026 and Conference-channel slices

### *7.2. Results*

Figure 4 plots Table 6 directly. Reading the results against the two questions this section set out to answer: first, does the guarantee hold empirically – `dominant_tactic` and `platform`'s marginal coverage (0.908, 0.925) both meet the 0.90 target. The sets differ in

Table 6: Split-conformal coverage at 90% target ($\alpha$=0.10), pooled over 5 repeats, $n\approx$1035 pooled test predictions per task. `family/tool` report mean per-label coverage and the stricter all-labels-jointly-covered rate. Set size is out of the task's class count ($K$) for `dominant_tactic/platform`, or out of 2 (per label) for `family/tool`. `family` uses TF-IDF with logistic regression here rather than its best embedding representation.

| Task | $K$ | Coverage | Set size | Yr≥26 | Conf. |
|---|---|---|---|---|---|
| Dominant tactic | 11 | 0.908 | 2.60/11 | 0.913 | 0.926 |
| Platform | 4 | 0.925 | 1.19/4 | 0.932 | 0.899 |
| Family (per-label) | 18 | 0.911 (joint: 0.446) | 0.93/2 | $0.487^{\text{joint}}$ | $0.572^{\text{joint}}$ |
| Tool (per-label) | 7 | 0.907 (joint: 0.650) | 0.92/2 | $0.640^{\text{joint}}$ | $0.752^{\text{joint}}$ |

how much they say: platform sets average 1.19 of 4 classes, while dominant-tactic sets average 2.60 of 11, which is the price of the target coverage for a model this weak. Second, the conditional-coverage stress test: all four multiclass slice numbers stay within about 3 points of the marginal rate. These slices are subsets of the same random splits, so they also appear in the calibration data; they test coverage on corpus subgroups, not under distribution shift. `family` and `tool`'s per-label coverage (0.91, 0.91) is close to target, but their *joint* (all-labels-simultaneously) coverage is far lower (0.45, 0.65). That is an expected consequence of accumulating misses across 18 and 7 binary calibrations, not a failure of Equation (7), which was never a joint guarantee. Both figures exceed what independent labels would give (about 0.19 and 0.51), so per-label misses are correlated. Joint coverage also varies more across slices than the multiclass rates do: on Conference channels it is 13 points above the marginal for `family` and 10 for `tool`. Per Section 7.1, all of the above describes agreement with $y^{\text{KA}}$.

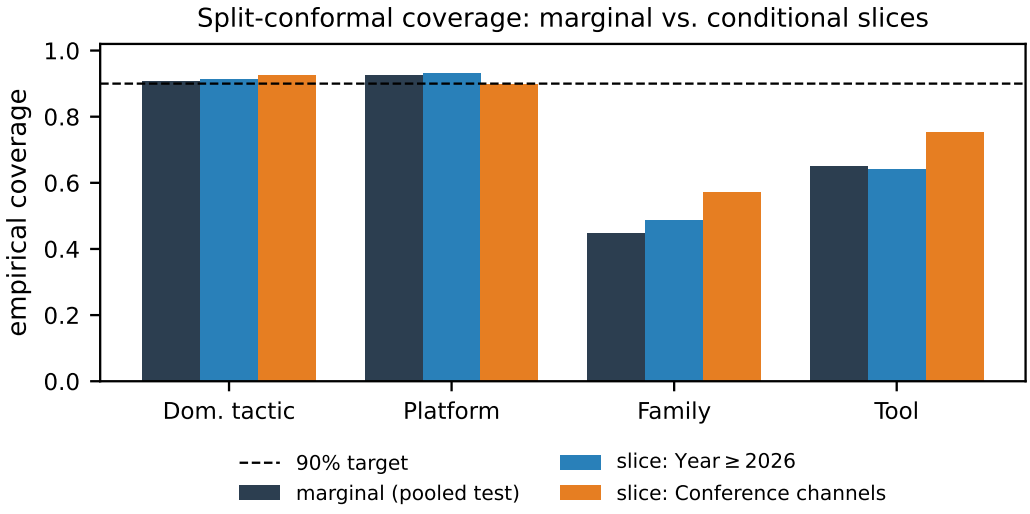



Figure 4: Empirical conformal coverage, marginal vs. the two conditional-coverage stress-test slices, against the 90% target (Table 6).

## 8. Out-of-Distribution Detection

OOD detection asks whether a model's own confidence or embedding-space behavior differs between in-distribution and out-of-distribution input. Scoring that needs no correct dominant-tactic label, since the question is which population an input came from. The weak labels are still involved: the shared classifier is trained on them, and the temporal and leave-rare-family setups define their populations with the regex's family list. Only the channel-type and external-corpus partitions are free of it.

### *8.1. Setups and methods*

Four setups, each with a stated in-distribution (ID)/out-of-distribution (OOD) split, drawn from real structure in the corpus rather than synthetic corruption:

1. **Temporal emergence.** ID: pre-2024 videos. OOD: videos from 2024 or later mentioning a family whose first appearance in this corpus is 2024 or later – verified directly against per-family first-occurrence year rather than assumed (Maze, Medusa, Black Basta, INC: 2024; Cactus: 2025), at Algorithm 3's assertion step. This is a family's first appearance in this corpus, not its real-world emergence.
2. **Channel-type shift.** ID: Vendor + DFIR channels. OOD: Conference + Independent channels.
3. **Leave-rare-family-out.** ID: videos never mentioning "Cl0p" (11 occurrences total – long-tailed, unlike Conti/LockBit/ Qilin/Akira, each with 19+). OOD: the 11 Cl0p-mentioning videos.
4. **External corpus.** ID: RansKnow transcripts. OOD: 20 Newsgroups [26] (`rec.sport.baseball` + `sci.space`, excluding 38 documents left empty after header, footer, and quote removal), included as a sanity check that the methods below can detect an obviously unrelated domain before trusting them on setups 1–3.

Four detection methods, computed from a shared `dominant_tactic` classifier (TF-IDF features, LightGBM – Table 3's best dominant-tactic result, and a natural multiclass fit for the softmax-family methods below), refit fresh on each setup's ID-only training pool so that no OOD signal leaks into fitting:

$$s_{\text{MSP}}(x) = 1 - \max_c \hat{p}(c \mid x) \tag{8}$$

$$s_{\text{energy}}(x) = -\log \sum_c \exp\big(f_c(x)\big) \tag{9}$$

$$s_{\text{maha}}(x) = \min_c \big(\phi(x) - \mu_c\big)^\top \Sigma^{-1} \big(\phi(x) - \mu_c\big) \tag{10}$$

$$s_{\text{conf}}(x) = 1 - \hat{p}\big(\hat{c}(x) \mid x\big) \qquad \text{(Equation (4) at the predicted class)} \tag{11}$$

Equation (8) is Hendrycks and Gimpel's max-softmax-probability baseline [17]. Equation (9) is the energy score [18], computed from LightGBM's raw pre-softmax margins $f_c(x)$ (*not* from $\hat{p}(c \mid x)$: probabilities already sum to 1, which makes Equation (9) degenerate to a constant if computed from them directly – a real pitfall caught while implementing this section, not a hypothetical one). Equation (10) is the class-conditional Gaussian Mahalanobis distance of Lee et al. [19], with $\phi(x)$ the 384-dim mean-pooled sentence embedding (Section 5's footnote)

and $\mu_c, \Sigma$ the ID-train class-conditional means and pooled covariance, using the classifier's own *predicted* classes to form groups (consistent with Section 4.3: no independently verified labels exist to group by instead). $\Sigma$ is a Ledoit-Wolf shrinkage estimate, because the temporal setup has only 162 training videos in 384 dimensions and its sample covariance is singular. Equation (11) reuses Section 7's LAC score, evaluated at the predicted rather than true class for OOD flagging – this is deliberately, not accidentally, the same quantity as Equation (8); the two are reported separately for comparability with Equation (5)'s calibrated threshold (does a point's own score exceed the 90%-target $\hat{q}$), which is the operationally distinct number.

**Algorithm 3** OOD setup evaluation

**Require:** corpus $\mathcal{D}$, ID/OOD partition rule for the setup

1: **assert** every claimed-emerging family's first-occurrence year matches the setup's stated cutoff ▷ caught a factual error at this step – see note below
2: split ID pool into $\mathcal{D}_{\text{train}}$ (75%) / $\mathcal{D}_{\text{test}}^{\text{ID}}$ (25%)
3: fit `dominant_tactic`/TF-IDF/LightGBM on $\mathcal{D}_{\text{train}}$
4: score $\mathcal{D}_{\text{test}}^{\text{ID}}$ and $\mathcal{D}_{\text{OOD}}$ under Equations (8) to (11)
5: report AUROC(ID vs. OOD) per method, plus the conformal exceedance-rate contrast

The assertion in Algorithm 3 is not decorative: the temporal- emergence setup's family list was originally drafted from memory as "Qilin, DragonForce, Black Basta," before checking actual first-occurrence years against the corpus. Qilin's first real occurrence is 2019, not post-2023, and DragonForce does not appear in this corpus at all – both would have silently produced a meaningless setup (there is no real ID/OOD separation if the "emerging" families already appear throughout the ID training period). The corrected list (Maze, Medusa, Black Basta, INC, Cactus) is the one actually used in Table 7 and is verified by Algorithm 3 on every run, not just checked once by hand.

### *8.2. Results*

Table 7: OOD-detection AUROC by setup and method (Equations (8) to (11)). $n_{\text{ID}}/n_{\text{OOD}}$ are ID-test/OOD sample counts. Stratified bootstrap 95% intervals for every cell are in `outputs/phase4_ood.json`.

| Setup | $n_{\text{ID}}$ | $n_{\text{OOD}}$ | MSP | Energy | Mahal. | Conf. |
|---|---|---|---|---|---|---|
| Temporal emergence | 55 | 31 | 0.425 | 0.341 | 0.220 | 0.422 |
| Channel-type shift | 156 | 353 | 0.600 | 0.582 | 0.672 | 0.578 |
| Leave-rare-family-out | 256 | 11 | 0.304 | 0.196 | 0.145 | 0.238 |
| External corpus (sanity) | 259 | 1152 | 0.622 | 0.477 | 0.999 | 0.627 |

Figure 5 plots Table 7. Three observations:

**The confidence-based methods (MSP, energy, conformal) fail the sanity check.** On `external_corpus`, MSP reaches only 0.622 (95% interval 0.573–0.672) and energy 0.477. Diagnosing *why* is informative: 99.0% of 20 Newsgroups documents are classified `None` (no dominant tactic detected) by the classifier, against 18.5% of the ID test set, and with *higher* average confidence (mean $\hat{p}_{\max}$: 0.922 OOD vs. 0.895 ID). A sparse, off-topic TF-IDF vector

and a tactic-free ransomware transcript both produce the same signature, confidently "no tactics detected," so the classifier cannot distinguish "nothing here" from "wrong domain entirely." This is a known blind spot of softmax/probability-based OOD detection in general [17], reproduced concretely here. These methods reach 0.58–0.60 on channel-type shift and fall below 0.5 on the other two setups.

**Mahalanobis distance on sentence embeddings passes the sanity check but inverts on the family-based setups.** It reaches 0.999 (95% interval 0.996–1.000) on `external_corpus` and 0.672 (0.618–0.723) on channel-type shift, where Conference/Independent channels differ detectably in embedding space; Section 8.3 shows that both figures are flattered and gives the more conservative values. On temporal emergence (0.220, 0.130–0.324) and leave-rare-family-out (0.145, 0.065–0.244) it is below chance with intervals that exclude 0.5, so these are not noise: OOD videos sit *closer* to the class means than ID videos do. The covariance estimate is not the cause, since the earlier pseudo-inverse estimate gives 0.223 and 0.145. We tested two explanations. The first is selection: every OOD video in these setups names a family, against 25% and 20% of their ID test sets. Restricting ID to family-naming videos ($n$=14, $n$=51) raises the scores to 0.353 (0.191–0.532) and 0.257 (0.119–0.417), so the temporal inversion is no longer distinguishable from chance, while the leave-rare-family inversion narrows but persists. The second is length: mean-pooling more chunks pulls long videos toward the class means (Spearman $\rho$ between distance and word count is $-0.66$ and $-0.71$), and OOD videos are longer (median 6,288 vs. 2,950 words and 6,480 vs. 5,036). Adjusting log distance for log word count leaves 0.256 and 0.148, so length does not explain the inversion by itself. The combined effect of the two remains untested, and with $n_{\text{OOD}}$=11 the leave-rare-family interval stays wide. Because both populations are selected through the regex family list, these two setups are a weak test of whether a detector notices novel families.

**The conformal exceedance-rate contrast (Equation (5)'s $\hat{q}$) is uninformative here by construction.** The calibrated threshold is at least 0.997 in every setup, because the classifier assigns the weak label almost no probability on more than a tenth of its calibration points ($n$ between 41 and 194). The OOD-flagging score is taken at the *predicted* class (Equation (11)), so it cannot exceed $1 - 1/K$, about 0.91 with 11 classes, and no point can cross the threshold. Equation (11)'s AUROC column is the more informative number from this method; the zero exceedance rates are in the released JSON (`outputs/phase4_ood.json`).

A single AUROC number per (method, setup) cell is a summary of a comparison between two full score *distributions*, and the two figures below show more of that picture. Figure 6 plots the MSP score density for the ID-test and OOD populations, per setup, as a 3D ridgeline: the blue (ID) and red (OOD) curves overlap heavily and both peak in the low-score, high-confidence region, the visual counterpart of MSP's weak AUROCs. Figure 7 is a 3D t-SNE projection of the same 384-dim sentence embeddings the Mahalanobis method (Equation (10)) computes its distances on, colored by ID/OOD membership. OOD points are scattered through the ID cloud in all three setups shown. t-SNE distorts distances, so this is illustrative only: it does not show the separation the Mahalanobis AUROCs measure on channel-type shift, and should not be read as evidence for or against it.

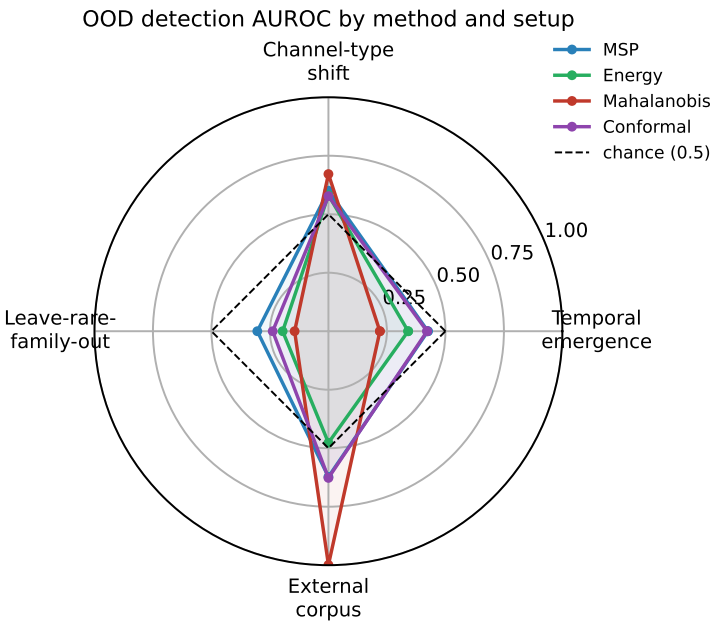



Figure 5: OOD-detection AUROC by method (color) and setup (axis), against the chance line at 0.5 (Table 7). Mahalanobis is the only method to clear chance by a wide margin, on the channel-type and external setups, and falls well below it on the other two.

### *8.3. Leakage and confound checks*

We audited these results for information shared between training and evaluation (`Scripts/phase4_leakage_audit.py`). No video or URL appears in both training and evaluation in any setup, and every fitted component (vectorizer, classifier, class means, covariance) uses training data only. Four indirect effects were tested.

*Shared channels.* The ID split is random by video, so 98–100% of ID test videos share a channel with training, while the OOD channels of channel-type shift never do, which favors separation. Holding ID channels out lowers its mean Mahalanobis AUROC over 5 seeds from 0.677 to 0.644 (range 0.608–0.688), so the 0.672 in Table 7 is modestly inflated but the effect survives. The inversions also survive: temporal emergence stays at 0.199, or 0.204 when OOD is limited to channels unseen in training, and leave-rare-family-out at 0.124. Every Cl0p video comes from a channel that also appears in training, so that setup cannot be held out on both sides.

*Duplicates.* Eight transcript pairs exceed 0.98 cosine similarity, including the four videos listed twice (Section 3). Dropping evaluation videos with a near-duplicate in training changes no Mahalanobis AUROC by more than 0.004.

*Unmatched family mentions.* Four of the 217 pre-2024 ID videos contain a term matching an emerging family that the context-gated regex did not tag. Removing them moves temporal emergence's Mahalanobis AUROC from 0.220 to 0.261, and no ID video contains a Cl0p term.

*Length and formatting in the sanity check.* The stored transcripts contain timestamp lines and have a median of 4,985 words, while the 20 Newsgroups posts have no timestamps and a median of 74 words, and Mahalanobis distance falls as documents get longer. Removing timestamps from the corpus and re-embedding it moves the sanity check only from 0.999

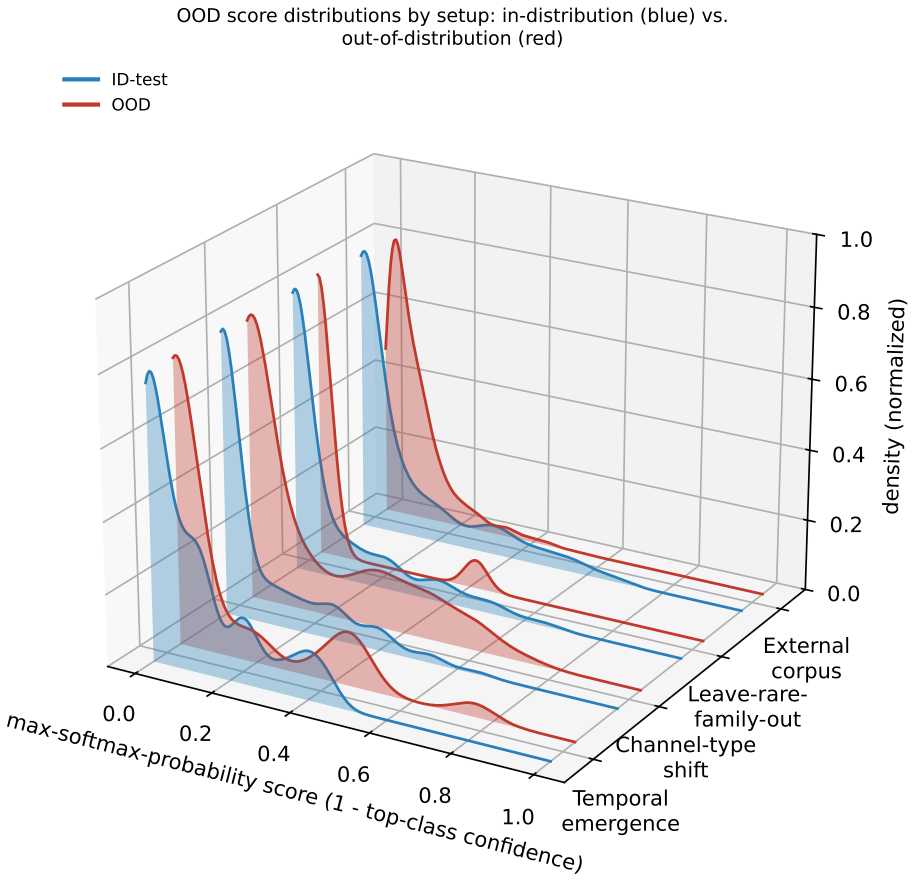



Figure 6: ID-test vs. OOD max-softmax-probability score density per setup, as a 3D ridgeline (kernel density estimate, not binned histograms). Overlap between the blue and red curves is the visual counterpart of MSP's weak AUROCs.

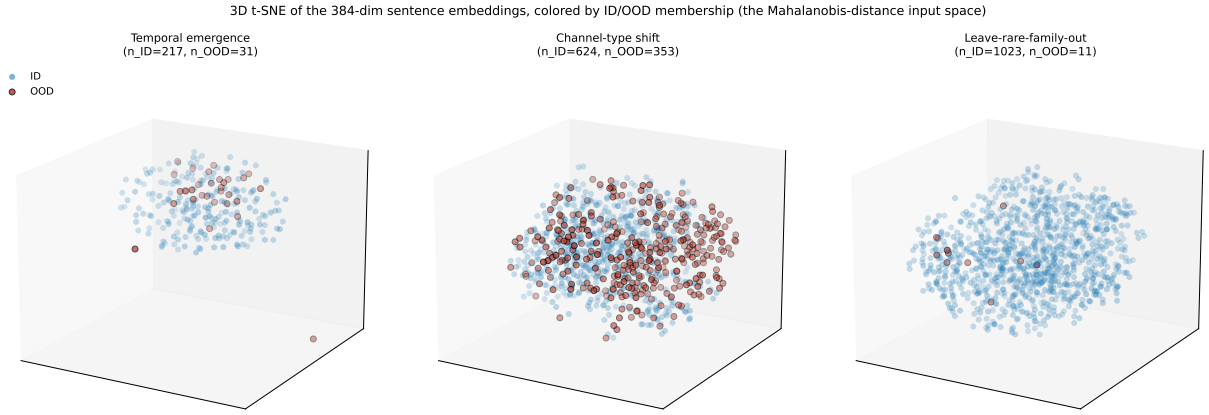



Figure 7: 3D t-SNE of the 384-dim sentence embeddings per setup, colored by ID/OOD membership, for the three within-corpus setups. Illustrative only, since t-SNE distorts distances: no setup shows visibly separated clusters.

to 0.996. Truncating each ID test video to its first 200-word chunk as well lowers it to 0.943 (0.930–0.955), which we take as the fairer figure. It is conservative, because the same truncation alone separates ID videos from their own full-length versions at AUROC 0.994. Timestamp removal moves the other Mahalanobis AUROCs by at most 0.022. Effects of timestamps on the TF-IDF baselines in Section 5 were not separately tested.

Taken together, Section 8.2 does not support a general claim that this pipeline's models reliably detect distributional shift. It supports a narrower one: softmax-confidence methods have an identifiable blind spot on this classifier (the "no signal" vs. "wrong domain" confusion above), embedding distance detects a change of domain and, more modestly, of channel type, and neither approach shows that it notices a video naming a family absent from training.

## 9. Interpretability and Perturbation Robustness

Unlike Sections 5–7, this section asks questions about model *behavior* – does it exploit shortcuts, what does it rely on, how much do its predictions move under semantically-irrelevant edits – none of which require independent ground truth to answer, so Section 4.3's scope note applies only where a model's training labels matter.

### *9.1. Confound checks*

**Length.** Transcript word count correlates moderately with `Tactic_Total_Mentions` (Pearson $r = 0.409$) and weakly with `Family_Count` ($r = 0.131$), but not at all with `Tool_Total_Mentions` ($r = 0.043$, not significant) – the confound is not uniform across signal types. Because it is mathematically guaranteed that dividing every tactic's count for one video by that video's own word count cannot change which tactic has the highest count (argmax is scale-invariant per row), the confound cannot affect how the `Dominant_Tactic` label itself was derived; the remaining question is whether it inflates a model given those counts. Re-scoring logistic regression on structured features with length-normalized (mentions per 1,000 words) instead of raw counts gives 0.753 macro-F1 against 0.764, a 1.1-point difference, not a length-exploitation artifact. This check covers only the structured representation, which Table 3 excludes for this task as circular; length is not tested for the TF-IDF model.

**Channel identity.** 486 of 1,034 transcripts (47%) contain at least one literal mention of their own channel's name. Masking those mentions and re-scoring TF-IDF LogReg under both CV protocols leaves the `channel_grouped`-vs-`stratified_random` gap unchanged for `family` (0.028 raw → 0.031 masked) and for `dominant_tactic` (0.020 raw → 0.029 masked). Whatever a model exploits when a channel is not held out, it is not the channel literally naming itself – the leakage this split protocol exists to catch is stylistic or topical, not addressable by name-masking.

### *9.2. Counterfactual explanations*

Motivated directly by Finding A: does a classifier trained on the *fixed* labels still show the regex's original word-collision fragility? TF-IDF+LogReg models were trained on today's corrected `Family` labels in 5 stratified folds, and every video was scored only by a model that did not train on it, because the videos in question are labeled not-`Play` in the training data. The test set is the 275 videos the Section 4.1 fix removed from `Play`, meaning the fix's own judgment rather than a per-video human review. Of these, 8 (2.9%) still reach $P(\text{Play}) \geq 0.5$,

with a maximum of 0.999. Their median probability is 0.005, against 0.001 for the 648 other not-`Play` videos, and the model ranks them above those videos with AUROC 0.58 (standard error about 0.02). That is a weak trace of the collision rather than either a clean bill of health or wholesale inheritance. It should be read against a weak model: only 18.9% of the 111 videos labeled `Play` reach 0.5, with a median of 0.019.

The complementary question – what drives the model's confident `Play` predictions – was then asked with a greedy word-*type* removal counterfactual search (remove the vocabulary word whose removal most reduces $P(\text{Play})$, repeat until the prediction flips or a budget is exhausted) on the 8 labeled `Play` videos with the highest held-out probability (all at least 0.97). `play` appears in 3 of 8 removal sets, and **0 of 8 cases flip on removing "play" alone**. Every case flips within 1–8 removals, mostly generic words rather than ransomware-specific vocabulary: the ASR filler "uh" appears in 5 of 8 sets, and one case flips on removing "uh" alone. So the model's confident predictions rest on a diffuse, easily disturbed set of words rather than one trigger word or a clean "ransomware vocabulary," a finding about the decision boundary that is separate from the word-collision question.

### *9.3. Perturbation robustness*

Untargeted counterpart to the counterfactual search above: two perturbation types, measuring flip rate (does the top prediction change) and mean absolute probability drift on freshly-fit TF-IDF+LogReg models. *Simulated ASR noise* (compound proper nouns split apart, e.g. `WannaCry`→"wanna cry," `Mimikatz`→"mimi cats," plus 8% random token deletion) was applied to the 74 official-caption videos specifically – the "clean" subset, simulating what those transcripts would look like if they had gone through the ASR cascade that produced the other 92.7% of the corpus (Finding B). *Synonym substitution* (common non-jargon words swapped for synonyms) was applied to a 150-video sample.

`dominant_tactic`, `platform`, and `relevance` are essentially unmoved by either perturbation (0.0–0.7% flip rate, mean drift $< 0.01$ in every case). `family` flips 20–26% of the time under both perturbations despite a much smaller mean drift ($\sim$0.002) than any other task – not a new problem, but the same weak-model issue from Table 3 (`family`'s best macro-F1 is 0.129) surfacing differently: closely-bunched, low-confidence probabilities mean a negligible nudge is enough to change which barely-ahead class wins. On the framing of "adversarial" robustness for this dataset: no real adversary submits transcripts to evade this classifier in production, so a full gradient-based adversarial-attack framework would be disproportionate to what this paper needs; the actual risk surface is ASR transcription noise and channel-style variation, which is what this sweep directly measures. Combined with Section 8's temporal-emergence and leave-rare-family-out setups, these results are the paper's answer to how the system handles evolving, novel, and adversarial-adjacent input – not three separate open questions.

## 10. Limitations and Ethics

This paper's single most consequential limitation is the absence of independent ground truth, discussed at length below; it opens this section as prose, not because the remaining limitations are unimportant, but because they are all of a smaller and more familiar kind (sampling bias, ASR noise, source rights, dual use) that most empirical NLP papers

carry in some form, whereas the ground-truth gap changes what every quantitative claim in Sections 5–7 *means*, not merely how confidently it can be stated.

- **No independent ground truth.** This is the paper's central limitation, not one bullet among several, and is why it is stated repeatedly rather than once: this project does not include a human-annotated evaluation set independent of the Knowledge Agent pipeline (Section 4.3). Recruiting external annotators would have required ethics approval beyond the project's existing approval for public data, and was not undertaken; a single-investigator annotation pass was considered and rejected as still too close to the system under evaluation to count as independent evidence. Every result in Sections 5–7 is consequently agreement with an audited but demonstrably imperfect extraction pipeline (Finding A), not verified accuracy. Section 8's OOD scores and Section 9's behavioral diagnostics are less directly affected, although two OOD setups and the `Play` check are defined using the regex's labels. Independent annotation, with whatever ethics review the authors' institution requires, is the single highest-value extension of this work. It is more valuable than any additional modeling in Sections 5–7 would be, since none of that modeling can currently be checked against anything but itself.

- **Sampling bias.** The corpus is conditioned on which channels operators chose to run, which videos they chose to title and describe in keyword-matchable ways, and YouTube's own recommendation and search surfacing – not a random or exhaustive sample of ransomware-relevant discourse.

- **ASR noise.** 92.7% of transcripts are machine-transcribed, mostly by `whisper_base`. Section 9.3's simulated-ASR-noise sweep bounds how much this moves predictions; Finding B's word-error-rate-on-jargon question (comparing ASR output against source audio directly, rather than a simulation of ASR-style errors) remains open.

- **Weak-label ceiling.** Every quantitative result in Sections 5–7 is bounded by the Knowledge Agent's own precision/recall, which Finding A shows is not perfect even after the disambiguation fix, and which Section 9.2 shows leaving a weak trace in at least one trained model.

- **Source rights and platform terms.** The transcripts derive from talks, webinars, and walkthroughs whose speakers and channel operators hold their own rights, and they were obtained through the YouTube Data API, `yt-dlp`, and audio download for Whisper transcription. The release contains transcript text and metadata but no audio or video. A video being public does not by itself grant a right to redistribute its speech as text, and we did not obtain permission from individual channels or speakers. YouTube's terms of service may also restrict automated collection of this kind, so re-users should check the terms that apply to them. The authors' CC0 dedication covers only the authors' own contributions (the extracted features, labels, and compiled metadata). It grants no re-use rights in the transcript text or the underlying talks, which remain with their owners and are shared here for research use. A rights holder can have a transcript removed from future releases by opening an issue on the repository or by contacting the corresponding author.

- **Dual use.** This is a corpus of ransomware tactics, tools, and case studies extracted from public educational content, and it is worth being direct about what that does and does not change. Every underlying video is already public, already indexed by YouTube's own search, and produced by defenders (vendors, DFIR practitioners, conference speakers) specifically to be watched by other defenders; RansKnow discloses no technique, tool, or tactic that is not already public, and the source material is materially less operationally detailed than existing public resources built for exactly this kind of aggregation – the MITRE ATT&CK knowledge base itself, and the CTI reporting datasets in Section 2 [8, 9], both already structure comparable tactic/technique/tool information at finer granularity and higher label quality than RansKnow's weak-label extraction achieves (Section 4.3). What RansKnow adds is not new operational detail but a different modality (transcribed spoken narration rather than written reports) and different scale accessible without specialized CTI tooling. The realistic marginal risk is a small reduction in the effort needed to *aggregate* already-scattered public information into one structured, searchable place, rather than any disclosure of non-public technique detail; this is the same tradeoff every public threat-intelligence aggregation resource makes; and the corpus's actual utility, given Section 4.3's weak-label ceiling and Section 5's weak family/tool classification performance, currently favors defensive research uses (weak-supervision methodology, corpus-level trend analysis) far more than it favors any operational offensive use, for which primary-source technical documentation would be a far more direct and reliable starting point than this corpus's noisy, ASR-transcribed, regex-extracted labels.

## 11. Conclusion

We presented RansKnow, an open MITRE ATT&CK-aligned corpus of 1,128 ransomware-related YouTube video records, 1,034 with transcripts. The audit shows that a plausible regex label can be dominated by detections a simple context check rejects: 71% of pre-fix `Play` detections (Table 1). A corpus released without such an audit would carry this error silently.

Against the audited labels, models reproduce relevance well (macro-F1 0.90) but dominant tactic poorly (0.38). In out-of-distribution detection, softmax methods fail a sanity check that Mahalanobis distance passes, yet Mahalanobis distance scores below chance when out-of-distribution videos are selected for naming a family. The audit and the counterfactual and perturbation checks can be reused on any dataset labeled by keyword or regex extraction over scraped or transcribed text.

The priority for future work is independent human annotation of a subset of the labels (Section 10). Until then, every quantitative result outside Section 8 describes agreement with the pipeline: a macro-F1 of 0.90 on `relevance` means a LightGBM classifier on TF-IDF reproduces the regex output at that rate, not that it identifies relevant videos at that rate.

## CRediT authorship contribution statement

**Henry Kabuye:** Conceptualization, Methodology, Software, Data curation, Formal analysis, Investigation, Writing – original draft. **Biju Issac:** Supervision, Writing – review & editing.

## Declaration of competing interest

The authors declare no competing interests.

## Data availability

The RansKnow corpus, the extraction and audit pipeline, and all experiment code are publicly available at https://github.com/Henrinnes/RansKnow and on Kaggle at https://www.kaggle.com/datasets/henrykabuye/ransknow-v1. The transcripts derive from third-party videos; Section 10 describes the rights that remain with their owners.


## Acknowledgements

This work was supported in part by the Pro:NE (Pronortheast) Network through a Grant by UK Research and Innovation and Office for Students (Grant Number: RDF22/ProNE/JHA) through a PhD studentship. The authors thank the CyberNets Research Group, Northumbria University, for their insightful discussions and useful feedback throughout this research project. The opinions, findings, conclusions, or recommendations expressed in this material are solely those of the authors and do not necessarily represent the views of the supporting team.


## Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used a generative AI coding assistant in order to write and debug analysis code, rerun and check experiments, and edit the manuscript text. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

## References


[1] B. E. Strom, A. Applebaum, D. P. Miller, K. C. Nickels, A. G. Pennington, C. B. Thomas, MITRE ATT&CK: Design and Philosophy, Technical Report MP 18-0944, MITRE Corporation, 2018.

[2] C. Beaman, A. Barkworth, T. D. Akande, S. Hakak, M. K. Khan, Ransomware: Recent advances, analysis, challenges and future research directions, Computers & Security 111 (2021) 102490. doi:10.1016/j.cose.2021.102490.

[3] H. Oz, A. Aris, A. Levi, A. S. Uluagac, A survey on ransomware: Evolution, taxonomy, and defense solutions, ACM Computing Surveys 54 (2022) 1–37. doi:10.1145/3514229.

[4] P. H. Meland, Y. F. F. Bayoumy, G. Sindre, The ransomware-as-a-service economy within the darknet, Computers & Security 92 (2020) 101762. doi:10.1016/j.cose.2020.101762.

[5] G. Husari, E. Al-Shaer, M. Ahmed, B. Chu, X. Niu, TTPDrill: Automatic and accurate extraction of threat actions from unstructured text of CTI sources, in: Proceedings of the 33rd Annual Computer Security Applications Conference, 2017, pp. 103–115. doi:10.1145/3134600.3134646.

[6] V. Legoy, M. Caselli, C. Seifert, A. Peter, Automated retrieval of ATT&CK tactics and techniques for cyber threat reports, arXiv:2004.14322, 2020.

[7] V. Orbinato, M. Barbaraci, R. Natella, D. Cotroneo, Automatic mapping of unstructured cyber threat intelligence: An experimental study, in: 2022 IEEE 33rd International Symposium on Software Reliability Engineering (ISSRE), 2022, pp. 181–192. doi:10.1109/ISSRE55969.2022.00027.

[8] S. K. Lim, A. O. Muis, W. Lu, C. H. Ong, MalwareTextDB: A database for annotated malware articles, in: Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), Association for Computational Linguistics, Vancouver, Canada, 2017, pp. 1557–1567.

[9] L. Lange, M. Müller, G. H. Torbati, D. Milchevski, P. Grau, S. C. Pujari, A. Friedrich, AnnoCTR: A dataset for detecting and linking entities, tactics, and techniques in cyber threat reports, in: Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024), 2024.

[10] M. Mintz, S. Bills, R. Snow, D. Jurafsky, Distant supervision for relation extraction without labeled data, in: Proceedings of the Joint Conference of the 47th Annual Meeting of the ACL and the 4th International Joint Conference on Natural Language Processing of the AFNLP, Association for Computational Linguistics, Suntec, Singapore, 2009, pp. 1003–1011.

[11] A. Ratner, S. H. Bach, H. Ehrenberg, J. Fries, S. Wu, C. Ré, Snorkel: Rapid training data creation with weak supervision, Proceedings of the VLDB Endowment 11 (2017) 269–282.

[12] C. Northcutt, L. Jiang, I. Chuang, Confident learning: Estimating uncertainty in dataset labels, Journal of Artificial Intelligence Research 70 (2021) 1373–1411. doi:10.1613/jair.1.12125.

[13] C. G. Northcutt, A. Athalye, J. Mueller, Pervasive label errors in test sets destabilize machine learning benchmarks, in: Proceedings of the Neural Information Processing Systems Track on Datasets and Benchmarks, 2021.

[14] V. Vovk, A. Gammerman, G. Shafer, Algorithmic Learning in a Random World, Springer, 2005.

[15] J. Lei, M. G'Sell, A. Rinaldo, R. J. Tibshirani, L. Wasserman, Distribution-free predictive inference for regression, Journal of the American Statistical Association 113 (2018) 1094–1111.

[16] A. N. Angelopoulos, S. Bates, A gentle introduction to conformal prediction and distribution-free uncertainty quantification, Foundations and Trends in Machine Learning 16 (2023) 494–591. ArXiv:2107.07511.

[17] D. Hendrycks, K. Gimpel, A baseline for detecting misclassified and out-of-distribution examples in neural networks, in: International Conference on Learning Representations (ICLR), 2017.

[18] W. Liu, X. Wang, J. D. Owens, Y. Li, Energy-based out-of-distribution detection, in: Advances in Neural Information Processing Systems (NeurIPS), 2020.

[19] K. Lee, K. Lee, H. Lee, J. Shin, A simple unified framework for detecting out-of-distribution samples and adversarial attacks, in: Advances in Neural Information Processing Systems (NeurIPS), 2018.

[20] H. Kabuye, RansKnow: A ransomware knowledge dataset, https://www.kaggle.com/datasets/henrykabuye/ransknow-v1, 2026. Accessed 2026-08-11.

[21] H. Kabuye, RansKnow: Dataset, extraction pipeline, and experiment code, https://github.com/Henrinnes/RansKnow, 2026. Accessed 2026-11-01.

[22] A. Radford, J. W. Kim, T. Xu, G. Brockman, C. McLeavey, I. Sutskever, Robust speech recognition via large-scale weak supervision, in: Proceedings of the 40th International Conference on Machine Learning, volume 202 of *Proceedings of Machine Learning Research*, 2023, pp. 28492–28518.

[23] N. Reimers, I. Gurevych, Sentence-BERT: Sentence embeddings using Siamese BERT-networks, in: Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP), 2019, pp. 3982–3992. doi:10.18653/v1/D19-1410.

[24] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, T.-Y. Liu, LightGBM: A highly efficient gradient boosting decision tree, in: Advances in Neural Information Processing Systems, volume 30, 2017.

[25] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, É. Duchesnay, Scikit-learn: Machine learning in Python, Journal of Machine Learning Research 12 (2011) 2825–2830.

[26] K. Lang, NewsWeeder: Learning to filter netnews, in: Proceedings of the Twelfth International Conference on Machine Learning, 1995, pp. 331–339. doi:10.1016/B978-1-55860-377-6.50048-7.

## Appendix A. Keyword List

The 21-term inclusion keyword list, matched case-insensitively against video title and description (Section 3), verbatim from `Scripts/fetch_transcripts.py`'s `KEYWORDS` constant:

```
ransomware, extortion, double extortion, encrypt, encryption, decrypt,
data leak, incident response, "ir", lockbit, alphv, blackcat, conti,
ryuk, cl0p, clop, akira, revil, sodinokibi, ransomware-as-a-service,
raas
```

The bare token "`ir`" is matched with surrounding spaces (`" ir "`) specifically to avoid matching it as a substring of unrelated words.